Lab 2: Frequency Distributions and Graphs

Covers Chapter 2: Frequency Distributions | 50-minute lab | Continues from Lab 1

Before you start: Open SPSS, then open your saved file from last week: **File** → **Open** → **Data** → navigate to where you saved it → select `Lab1_YourLastName.sav` → **Open**. You should see your 10-row, 6-variable dataset (StudentID, Major, ClassYear, StudyHours, NetflixHrs, Stress). We are *not* creating new variables today — we're analyzing the ones you already built.

Part 1: Frequency Table + Bar Chart for a Categorical Variable (13 min)

We'll start with **Major** — a nominal variable — and build both a frequency table and its graph in one pass.

1. **Analyze** → **Descriptive Statistics** → **Frequencies**
2. Click **Major** in the left-hand list, click the arrow to move it into the **Variable(s)** box.
3. Click the **Charts...** button.
   - Select the **Bar charts** radio button.
   - Under "Chart Values," select **Percentages** (instead of the default "Frequencies") — this shows the proportion of students in each major, not just the raw count.
   - Click **Continue**.
4. Click **OK**.

Two things appear in the Output Viewer:

A frequency table. Match its columns to the textbook's vocabulary:

SPSS column	Chapter 2 term
Frequency	f — the raw count in each category

SPSS column	Chapter 2 term
Percent	% — same as computing p × 100, where p = f/N
Valid Percent	% excluding any missing data (identical to Percent here, since we have none)
Cumulative Percent	running total as you move down the table — this is the basis for percentile ranks

A bar chart. Notice the spaces between the bars — that's deliberate. Chapter 2 draws bar charts with gaps specifically for nominal/ordinal data, to visually signal that the categories are separate and unordered, unlike a histogram's touching bars for numerical data.

> Checkpoint
>
> In your output, which major has the highest frequency? What percentage of the class does it represent?

Part 2: Independent Practice — ClassYear (5 min)

Repeat the same process for **ClassYear**, an ordinal variable, on your own:

1. **Analyze** → **Descriptive Statistics** → **Frequencies** → move **ClassYear** in → **Charts** → **Bar charts** → **Percentages** → **Continue** → **OK**.

> Checkpoint
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> Looking at the Cumulative Percent column, what percentage of students are Sophomores or below? (This is exactly what a percentile rank tells you — the percentage of the group at or below a given point.)

Part 3: A Numerical Variable — and Why It Gets Messy (8 min)

Now let's try the same recipe on a **scale (numerical)** variable: **StudyHours**.

1. **Analyze** → **Descriptive Statistics** → **Frequencies**

2. Remove Major/ClassYear if they're still listed; move **StudyHours** in instead.
3. Click **Statistics...** and check the box for **Quartiles**. Click **Continue**.

    - Quartiles report the 25th, 50th, and 75th percentiles — this is SPSS computing exactly the percentile concept from this chapter.

4. Click **OK** (leave Charts unchecked for now).

Look at the frequency table. With only a few exceptions, every row has a frequency of 1 — because with continuous data spread across a wide range, individual scores rarely repeat.

This is precisely the problem Chapter 2 describes as the reason for **grouped frequency distribution tables**: when scores cover a wide range, listing every individual value produces a long, uninformative table instead of a picture you can actually interpret.

> Checkpoint
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> Look at the Percentile Values table in your output. What StudyHours value corresponds to the 50th percentile?

Part 4: Building a Grouped Frequency Table (12 min)

We'll now group StudyHours into class intervals, following the textbook's guidelines: about 3–10 intervals (we have a small dataset, so we'll aim for 3), a simple interval width (5 is a good round number), and a bottom score that's a multiple of the width.

Why the boundaries below look slightly unusual: we're setting the cut points at the *real limits* (4.5, 9.5, 14.5, 19.5) rather than the whole numbers themselves. This matters because our data includes decimals (like 9.5 hours) — using the real limit means every possible score falls into exactly one interval with no gaps, exactly as Chapter 2 describes.

1. **Transform → Recode into Different Variables**
2. Click **StudyHours**, move it into the middle box (it will show as `StudyHours --> ?`).
3. In the **Output Variable** box on the right: type `StudyGroup` in **Name**, type `Study hours (grouped)` in **Label**. Click **Change**.
4. Click **Old and New Values...** A new dialog opens. Set up three ranges, clicking **Add** after each:

    - Old Value → select **Range**: `4.5` through `9.4`. New Value: `1`. Click **Add**.
    - Old Value → select **Range**: `9.5` through `14.4`. New Value: `2`. Click **Add**.
    - Old Value → select **Range**: `14.5` through `19.4`. New Value: `3`. Click **Add**.

5. Click **Continue**, then **OK**. A new column, `StudyGroup`, appears at the end of your Data View.
6. Switch to **Variable View**. Find the `StudyGroup` row and finish setting it up like you did in Lab 1:

   - **Decimals:** `0`
   - **Values:** 1 = `5-9 hrs`, 2 = `10-14 hrs`, 3 = `15-19 hrs`
   - **Measure:** select **Ordinal** (the groups now have a meaningful order, even though the underlying variable is scale)

7. Run **Analyze** → **Descriptive Statistics** → **Frequencies** on **StudyGroup**.

Compare this table to Part 3's. Same underlying data, but now it tells a story you can actually see at a glance — that's the entire purpose of grouping.

Part 5: Histogram and Shape (6 min)

1. **Analyze** → **Descriptive Statistics** → **Frequencies**
2. Move **StudyHours** (the original, ungrouped variable) into the Variable(s) box.
3. Click **Charts...** → select **Histograms** → click **Continue** → **OK**.

SPSS automatically bins the continuous variable for you (similar in spirit to what you just did by hand in Part 4). Look at the shape of the bars.

> Checkpoint — answer using your histogram
>
> - Is the distribution roughly **symmetrical**, or does it look **skewed**?
> - If skewed, which direction is the tail pointing — toward the higher scores (positive skew) or the lower scores (negative skew)?
> - Take a look at where SPSS started the Y-axis. Does it start at zero? (Keep this in mind — we'll come back to it below.)

Take-Home Lab Report (due before next class)

1. **By-hand check.** Using the Major frequency table from Part 1, compute ΣfX by hand — but first you'll need to assign a numeric code to each major (use the same codes from your Values labels: 1=Psychology, 2=Biology, 3=Business, 4=Undecided) and treat frequency $\times$ code as fX for each row. Show your work. (This is an artificial calculation since Major

is nominal and its codes aren't truly quantitative — that's the point: explain in one sentence why ΣfX is mathematically possible here but not meaningful.)

2. **Polygon by hand.** Using your Part 4 grouped frequency table (StudyGroup), sketch a frequency polygon on paper: place a dot above the *midpoint* of each interval at a height equal to its frequency, connect the dots, and drop the line to zero one interval-width beyond each end. (Recall: the midpoint of a class interval is the average of its highest and lowest apparent limits.)
3. **Stem-and-leaf by hand.** Construct a stem-and-leaf display for the 10 StudyHours values. What advantage does this display have over the grouped frequency table you built in Part 4?
4. **Population vs. sample.** Our 10 students are a sample. If we had data on the *entire population* of students at this university instead, would we typically graph actual frequencies or relative frequencies? Why does the textbook recommend relative frequencies for very large groups?
5. **The Y-axis question.** Referring to Box 2.1 in the textbook (the homicide-rate graph example) and what you noticed at the end of Part 5: why does it matter whether a graph's Y-axis starts at zero? Give a one-sentence explanation of how a manipulated Y-axis could mislead someone looking at your StudyHours histogram.
6. **Percentile interpretation.** In your own words, explain what it means that a StudyHours score has a percentile rank of 70%. Use the cumulative percent column from one of today's frequency tables as your example.